Operating Systems

Subject: Computer Science | NoteShare Library

OPERATING SYSTEMS

1. PROCESSES
New, Ready, Running, Waiting, Terminated
PCB: PID, PC, registers
Process vs Thread

2. SCHEDULING
FCFS: convoy
SJF: optimal
SRTF: preemptive
RR: quantum
Priority: aging
MLFQ

3. SYNCHRONIZATION
Critical section
Mutual exclusion
Peterson’s, Test-and-Set
Semaphore, Mutex, Monitor
Producer-Consumer, Dining Philosophers

4. DEADLOCK
4 conditions
Prevention, Avoidance, Detection, Recovery
Banker’s algorithm

5. MEMORY
Paging, Segmentation, Virtual
Page replacement: FIFO, LRU, Optimal, Clock
Thrashing

6. FILE SYSTEMS
Contiguous, Linked, Indexed
B-Tree directories
Permissions

7. I/O
Programmed, Interrupt, DMA
Disk scheduling: FCFS, SSTF, SCAN
RAID levels